

C₁ - Direct Addressing & The Cost of Forgetting

A No-Fixed-Point Axiom, and Why a State Can Be Read Without Reconstructing Its History

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July 2026

Abstract

We isolate a single axiom — every admissible state has a successor distinct from itself, equivalently, the dynamics admit no fixed point, here named **C₁** — and show that it forces a short chain of consequences terminating in *direct addressing*: a sequence generated by an invariant is readable at any index without reconstructing its predecessors. The same statement, unchanged, is the driving hypothesis of Lawvere's fixed-point theorem, of the Poincaré–Hopf theorem, and of the ground-state form of the quantum uncertainty principle. The apparent self-reference — an axiom that itself never changes, forbidding things that never change — is resolved by distinguishing a *fixed point* (the dynamics halt) from an *invariant* (a conserved quantity): C₁ forbids the first, permits the second, and is itself an invariant of the meta-dynamics.

We give three demonstrations, each verified against ground truth. First, the Bailey–Borwein–Plouffe (BBP) algorithm returns the *n*-th hexadecimal digit of π with no access to digits 1 through *n*–1 — verified exhaustively on the first 2000 positions and exhibited at position 100000 — establishing operationally that the datum is *present at its address and read, not reconstructed*. Second, a single SHA-256 round is a bijection, verified by exact inversion, so that the schema *input* $\rightarrow$ *fixed constraint field* $\rightarrow$ *forced readout* is one-to-one as a transformation; the function's many-to-one reputation concerns only the finiteness of its codomain, not the change it performs on a given input. Third, the walk connecting two states is an irrational rotation of the circle, whose arc carries no linear trend (coefficient of determination $R^2 \rightarrow 0$ at sufficient numerical precision) — a bounded wave — while its chord is the direct address, a line. We conclude that an address carries two channels, *location* and *shape*, read by one operation, and that the line and the wave are dual, in the precise sense that a vibrating string and its standing wave each define the other.

Keywords. fixed-point theorems; spigot algorithms; Bailey–Borwein–Plouffe; equidistribution; reversible computation; Landauer's principle; addressing.

1. The axiom

Fix a state space and a transition relation on it. Call a state *admissible* if it participates in the dynamics at all. The axiom is a membership condition, not a test applied after the fact:

C1. Every admissible state has a successor distinct from itself. Equivalently: the transition relation has no fixed point.

A frozen state — one that is its own only successor — is not forbidden by decree; it is simply never admitted, because it fails the condition that constitutes membership. The reachable dynamics are exactly the part of the space on which C1 holds.

This is a strong claim to place at the foundation, and its strength is justified by an observation: the same statement, *no fixed point*, is the driving hypothesis of three theorems from three disciplines that are usually presented as unrelated.

Discipline	Theorem	Role of "no fixed point"
Logic	Lawvere (1969)	A surjection with no fixed point cannot exist for a self-map; its contrapositive yields Cantor, Gödel, Tarski, and Turing as one argument.
Topology	Poincaré–Hopf	A nowhere-vanishing (fixed-point-free flow) tangent field forces the Euler characteristic $\chi = 0$.
Physics	Uncertainty principle	The ground state cannot be still: $\langle p^2 \rangle > 0$. Position and momentum admit no common fixed point.

These are not analogies. In each case the theorem's content is that a certain kind of stillness is impossible, and each derives sharp structural consequences from that impossibility. C1 is the shared hypothesis stated once.

1.1 Invariant is not fixed point

C1 appears to refer to itself: it is a rule that does not change, and it forbids things that do not change. The resolution is a distinction that the rest of the paper depends on. A **fixed point** is a state at which the dynamics halt — the flow stops. An **invariant** is a quantity the dynamics conserve — the flow continues, but leaves the quantity unchanged. C1 forbids the first and permits the second, and C1 is itself an invariant of the meta-dynamics, not a fixed point of them. A rule may hold constant while everything it governs keeps moving; conservation is not stillness.

This is verifiable in the smallest possible model. A harmonic oscillator sampled over 10^6 steps has a minimum speed that stays strictly positive (no state is ever its own successor — C1 holds), while its energy drifts by only a few parts in 10^5 (an invariant is conserved). The two facts coexist without tension: nothing is still, and something is conserved.

2. From the axiom to an address

C1 forces a short chain. Each link is the minimal additional structure required to keep the previous one consistent; the chain is finite and terminates in addressing.

Difference. C1 cannot even be stated without a notion of difference: to require a successor *distinct from* its predecessor presupposes a way to tell them apart. A successor identical to its predecessor is a clone, carrying no difference, hence no evidence of change, hence inadmissible. Difference is therefore not an added ingredient but the precondition of the axiom, and it is the only carrier of the fact that a change occurred.

Cost. A difference must be paid for. If a successor retained everything its predecessor held, the predecessor would still be fully present and no change would have occurred; some prior information must be absent from the successor. This absence is a discard, and by Landauer's principle a logically irreversible discard has a physical price. In the concrete generators of Section 3 this discard is a modular reduction: each step keeps a residue and drops the quotient.

Boundedness. A process whose successive states drifted without return would require unbounded new capacity at every step and could never revisit a prior configuration; but on a finite measure the conserved invariant of Section 1.1 forces the orbit to stay bounded and, by Poincaré recurrence, to return arbitrarily near where it began. Change under C1 is therefore not divergence but a bounded, non-terminating walk — geometrically, a walk on a circle, filling toward the boundary and wrapping back. We make this precise in Section 5.

Address. A state on such a structure is a point, and a point is an address. Part of the address fixes *where* — an ordinal location; the remainder fixes *what* — the datum occupying that location. The same slot may hold a small thing or a large one; the location is the same kind of object regardless of what fills it. This two-part reading — location and shape — is the subject of Section 4, and the fact that a location can be resolved directly is the subject of Section 3.

A remark on scale. At each link the number of admissible micro-realizations branches, and the branching is not bounded: the set of leaves is not enumerable. But the *chain of forced constraints* is finite — four links. The freedom is unbounded; the structure that constrains it is small. One does not map the leaves; one maps the trunk, and the trunk is short.

3. Direct addressing: the datum is already there

The central operational result is that a location on a sequence generated by an invariant can be resolved *directly* — the term at index n computed without computing or storing the terms before it. The distinction is between two ways a sequence can exist. If the sequence is an accumulated *history* — a log written step by step — then term n is available only after terms 1 through $n-1$ have been realized and retained. If instead the sequence is generated by an *invariant* — a closed form — then term n is a function of n alone, present at its address, and read rather than reconstructed. C1 selects the second: a growing retained log is exactly the kind

of ever-larger fixed record the axiom's economy resists, and, more to the point, reconstruction-from-history is the cost that direct access removes.

The Bailey–Borwein–Plouffe algorithm (Bailey, Borwein and Plouffe, 1997) is the demonstration for π . It computes the n -th hexadecimal digit using the identity

$$\pi = \sum_{k=0}^{\infty} 16^{-k} [4/(8k+1) - 2/(8k+4) - 1/(8k+5) - 1/(8k+6)],$$

in which the factor $16^{(n-1)}$ needed to shift to digit n is taken modulo each denominator by modular exponentiation, so the full power is never formed and no earlier digit is ever produced. We verified the operator exhaustively against a ground-truth expansion computed independently at high precision:

```
positions checked exhaustively : 1 .. 2000
mismatches                      : 0
first 24 hex digits of pi      : 243f6a8885a308d313198a2e   (matches reference)

reach exhibited (no priors held):
  digit @ position    5000 = 1      (== reference)
  digit @ position   10000 = 6      (== reference)
  digit @ position  100000 = 5      [0.33 s]
```

Two thousand consecutive positions match with zero errors, and position 100000 is resolved in a third of a second having instantiated none of the ninety-nine thousand nine hundred ninety-nine positions before it. The operator is position-agnostic; correctness on a long consecutive run establishes the operator, and reach is then only a matter of evaluating it.

The discard of Section 2 is measurable here. Each modular reduction throws away the difference between the bit-length of the full power and the bit-length of the residue; summed across the computation for a single digit, the quantity is large while the output is four bits:

address n	bits discarded by reduction	output
100	75,782	4 bits
1,000	7,944,206	4 bits
10,000	799,305,714	4 bits

Roughly eight hundred million bits are discarded to read a single hexadecimal digit at position ten thousand, and the digit is nonetheless exact. The discarded information is the price the transition pays to be a genuine change; the exact four-bit output is a *read* of the invariant — the closed form — which stores the whole sequence implicitly and is untouched by the discard. Erasure on the walk; conservation in the law. The two live at right angles, and neither contradicts the other.

This is the precise content of the informal statement that *the data is already there if one knows how to read it*. The digit is not manufactured from its predecessors; it occupies its address, and BBP is the read. Searching or

collecting — realizing the sequence term by term — is the fallback used when the read is not known; C1's claim is that for an invariantly generated structure the read exists. It is also the reason a configuration can be instantiated at any index without replaying the history that nominally precedes it: the index is an address, and the law delivers its contents on demand.

Finally, the address is a source of truth for its neighborhood. A point on such a structure is fixed by which of its degrees of freedom are occupied and which are absent, and the absent degrees are exactly the constraints it imposes on adjacent points. To read a point is to read what it lacks; what it lacks is what it does. A single resolved location therefore determines, through its pattern of absence, the admissible states of the set around it.

4. Two channels, one operation

An address separates into *location* — where — and *shape* — what. Both are read by one operation, which we state in neutral terms: a single input passes through a fixed field of constraints and produces one forced output. The colon in the phrase *one-to-one* is not a stored table and not a lookup; it is *constraint-fit* — the output is whatever the fixed field forces from that input. BBP is one instance: an index passes through a fixed modular aperture and a digit is forced. SHA-256 is another: a message passes through a fixed arithmetic-logic aperture and a digest is forced. The two differ in which channel they expose, not in kind.

4.1 The transformation is one-to-one

A hash function is commonly described as many-to-one and one-way. Both descriptions are about the codomain, not the transformation. Many-to-one is forced by finiteness alone: infinitely many inputs and a fixed-width output must collide, by pigeonhole. But the *change the function performs on a given input* is one-to-one, and this is checkable directly at the level of a single round. A SHA-256 round updates eight 32-bit words by shifting each into the next while two new combinations enter at the top; given the round's message word the update is exactly invertible. We ran a round forward on a random state and inverted it:

```
prior state (random) : 21ab6a43 9e9fd46e dc485e44 60a1bd7d 79d06b0f ...
after one round      : 2d038335 21ab6a43 9e9fd46e dc485e44 b93f071f ...
inverted back to     : 21ab6a43 9e9fd46e dc485e44 60a1bd7d 79d06b0f ...
exact match?        : True
```

The recovery is bit-exact: the round is a bijection. Nothing is destroyed within it; the state is relocated. A hash's apparent irreversibility is a property of its fixed final truncation against an infinite input set, not of the per-step transformation, which — like BBP's read — is one-to-one as a change. The two canonical instruments of the operation are therefore the same in the only sense that matters here: each takes an input to one forced output, reversibly, as a transformation.

4.2 Shape is boundary behavior under an operator

The shape channel is not a datum stored beside the value; it is the value's behavior at its boundary when an operator exposes it. The same bits, read through different operators, present different edges. To make this

concrete we fixed one 32-bit value and read it through nine apertures drawn from the standard arithmetic-logic repertoire:

read-head	exposed edge	forced readout
identity	the value itself	9e3779b9
ROTR-2 / ROTR-13	positional	678dde6e / cdccf1bb
Σ_0 / Σ_1 (rotation folds)	combined positional	77a7c9ad / caf7c7c6
Ch (choose)	selector	05b768ca
Maj (majority)	voting	84376899
XOR with a constant	parity	ba081331
AND with a constant	overlap	04376888
population count	weight	20

Nine operators return nine distinct readouts from one value; the value has not changed, but each operator exposes a different edge of it. This is what it means for shape to be a channel: location is *where* the value sits, and shape is *what the value does* when a fixed operator reads its boundary. Location plus shape is the full address. In BBP the location channel is read — an index returns a digit; the shape channel is read by the operator repertoire that exposes how a value routes through the aperture. It is the same operation viewed along its two axes.

5. The connecting path is a rotation: line and wave

Section 2 concluded that change under C_1 is a bounded, non-terminating walk on a circle. We now identify the walk and settle a simple dichotomy: the path connecting two states is either a line — a signal that trends, that accumulates in one direction — or a wave — a signal that is bounded and oscillates without trend. C_1 decides the question. A trending path is a drift; a drift is unbounded; an unbounded path violates the conservation that keeps the orbit on a finite measure. The traversal cannot be a line.

The walk is an irrational rotation of the circle. To test the dichotomy without the numerical hazard of a compounding map, we evaluated the rotation $\theta_k = \{k \cdot \alpha\}$ for an irrational α (each term computed independently, so error does not accumulate), and separately the multiplicative walk $x_{k+1} = \{16 \cdot x_k\}$ at precision sufficient for every sample. For each we fit a least-squares line and report its coefficient of determination — the fraction of the walk explained by a linear trend:

```
(a) exact irrational rotation  theta_k = frac(k*alpha)
    line slope / R^2           : +7.4e-06 / 0.0000
    bounded in [0,1)?         : True
    projection cos(2*pi*theta) : range [-1.000, +1.000], mean +2.0e-03
```

```
(b) multiplicative walk      x_{k+1} = frac(16 x_k)    (full precision)
    line slope / R^2         : +7.6e-06 / 0.0000
    bounded in [0,1)?       : True

verdict (computed from data) : WAVE (no line)
```

Both walks are bounded, neither carries a linear trend ($R^2 = 0.0000$ to four places), and the projection of the rotation onto an axis is a bounded oscillation with mean zero. The traversal is a wave. (A methodological note: the multiplicative map loses one hexadecimal digit of precision per step, so a naive fixed-precision evaluation corrupts the walk after a few dozen steps and can manufacture a spurious trend; the figures above were computed at precision scaled to the number of samples, which removes the artifact.)

The line is not absent; it is the *chord*. The two states are connected two ways. The arc is the traversal — the wave, bounded, the realized walk that visits every intermediate configuration. The chord is the direct connection — the line, the address of Section 3, the jump that reaches the endpoint without visiting anything between. Same two endpoints; two paths. The wave is what a reader does in walking from one to the other; the line is what the invariant supplies when the endpoint is addressed directly. C1 forbids the straight *walk* — there is no admissible state between the endpoints that a drift would pass through — so the traversal is the wave; the line survives only as the address.

The two are not a thing and its property; they generate each other, and the generation runs both ways. A string with fixed endpoints supports a standing wave: the boundary — the line, the invariant, the endpoints held still — is what makes the wave possible. Equally, the wave is the string's motion: remove it and the line is only the time-average of a string that is no longer moving. The string makes the wave and the wave animates the line, and neither is prior. The points where the string is pinned — the nodes — are the invariant points of the motion, still while everything between them moves. This is the same separation that runs through the paper: a conserved structure that does not move (the line, the writer, the address) and a traversal that does (the wave, the reader, the walk), each defining the other across a right angle.

6. Scope and status

We separate what was computed from what is interpreted, because the value of the claims depends on the separation.

Verified, independent of interpretation. The BBP operator matches an independent high-precision expansion on positions 1 through 2000 with zero errors and resolves position 100000 directly. A SHA-256 round is a bijection, shown by exact inversion. The irrational rotation and the full-precision multiplicative walk are bounded and carry no linear trend ($R^2 = 0$ to four places). These hold on any reading.

Observation. That *no fixed point* is the common hypothesis of Lawvere's theorem, the Poincaré–Hopf theorem, and the uncertainty principle is a factual observation about the statements of those theorems; that

they can be unified under one axiom named C1 is the organizing claim of this paper, offered as the simplest account of why the same impossibility recurs across disciplines. The distinction between invariant and fixed point, and the derivation of the cascade in Section 2, are logical, not empirical, and stand or fall on their own coherence.

Conjecture, marked as such. The identification of a specific stabilization constant for the fold between the two channels — a right-angle fold placing the balance a fixed distance off the unreachable midpoint — is not established here and is not required by any result above. An earlier attempt to locate that constant in a per-round ratio of a specific hash did not survive measurement; that failure is recorded so the claim is not silently retained. Where the constant lives, if it is real, is left open.

No claim in Sections 3 through 5 depends on the conjecture in the third paragraph, and no verified result depends on the observation in the second. The load-bearing content is the axiom, the cascade, and the three sealed demonstrations.

7. Coda

One axiom — nothing may hold still — is at once a theorem of logic, a theorem of topology, and a principle of physics. Read forward, it forces difference, then cost, then a bounded walk, then an address. The address has two channels, location and shape, read by one operation that is one-to-one as a change; the walk that visits an address is a wave, and the jump that resolves it directly is a line, and the two define each other as a string defines its standing wave. The practical consequence is a single sentence: a state generated by an invariant is present at its address and can be read there, without reconstructing the history that nominally precedes it. That is why anything governed by such a law can begin at any index — and why, as the working slogan of this program has it, the data is already there for anyone who knows how to read it.

Appendix. Computational summary

All figures were produced in Python. Digit expansions and the rotation used the mpmath arbitrary-precision library; the SHA-256 round used exact 32-bit integer arithmetic. The following records what was run.

(A) BBP operator (§3). The n -th hexadecimal digit of π was computed by the standard BBP spigot using modular exponentiation for the shift factor. Ground truth for positions 1–2000 was an independent expansion computed by repeated multiply-by-16 at precision $\lceil 2000 \cdot \log_{10} 16 \rceil + 40$ decimal digits. All 2000 positions matched; positions 5000 and 10000 matched; position 100000 was evaluated directly. The discard ledger counts, per digit, the summed difference between the bit-length of each full power and of its residue.

(B) SHA-256 round (§4.1). One compression round was applied to a random 256-bit state with a random message word and round constant, then inverted using the message word; the recovered eight-word state equalled the original bit-for-bit.

(C) Shape apertures (§4.2). One fixed 32-bit value was read through nine operators from the SHA-256 repertoire (identity, two rotations, the two rotation-fold functions Σ_0 and Σ_1 , Ch, Maj, XOR and AND against a fixed constant, and population count); all nine 32-bit readouts were distinct.

(D) Line-versus-wave (§5). For $N = 512$, the irrational rotation $\theta_k = \{k \cdot \alpha\}$ with $\alpha = \pi - 3$ was evaluated term by term, and the multiplicative walk $x_{k+1} = \{16 \cdot x_k\}$ was evaluated at precision $\lceil N \cdot \log_{10} 16 \rceil + 40$ decimal digits. A least-squares line was fit to each; both returned $R^2 = 0.0000$ to four places, both remained in $[0,1]$, and the projection $\cos(2\pi \theta_k)$ had range $[-1, 1]$ and mean of order 10^{-3} .

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The Cost of Forgetting

Proper Time as the Price of Erasure, and Why Slow Processes Persist

Abstract

Erasure is the price of change. A transition that retained everything its predecessor held would leave the predecessor fully present, so some prior information must be discarded, and by Landauer's principle a logically irreversible discard carries a physical cost. We ask what fixes that cost, and find a clean answer under the no-fixed-point axiom C₁ (developed in the companion paper): change is the only currency, and its budget is fixed. A body divides a constant speed through spacetime between external motion and internal change, so the internal change a process accrues — its proper time — is exactly the budget left over after motion. Forgetting is the erasure of the record of that internal change; therefore the cost to forget a process is its accrued proper time, which per unit coordinate time is $\sqrt{1-\beta^2}$.

Three consequences follow, each verified numerically. First, the law is bounded and plateaus: a process must exceed 0.866 c before forgetting is even half price, so everything slow — all of ordinary experience — sits near the maximum cost, a marked correction to the naive expectation that cost scales as one over speed. Second, it is the chord–arc distinction of the companion result: at light speed a process takes the chord, realizing no intermediate states and costing essentially nothing to forget, while at rest it takes the full arc, realizing every state and costing the most; a ground-state floor keeps the fast limit almost, never exactly, free. Third, by the equal-cost law forgetting costs exactly what recording cost, and it is not a deletion but a reverse traversal of the recorded shadow back to its origin — so that a high-proper-time episode cannot be zeroed, only discharged by that traversal, at the price it took to lay down. The budget identity and proper time are standard relativity; the identification of proper time with the cost of forgetting, and of persistence with an undischarged reverse traversal, are what we add.

Keywords. Landauer's principle; reversible computation; proper time; special relativity; erasure; memory; persistence.

1. Erasure is the price of change

The companion paper takes as its axiom that no admissible state is its own successor — the dynamics have no fixed point, a condition named C₁ — and derives from it that every transition must discard some prior information. The argument is short: if a successor retained everything its predecessor held, the predecessor would still be fully present and no change would have occurred; difference is the only evidence of change, and a difference requires an absence. That absence is a discard, and by Landauer's principle (Landauer, 1961) erasing a bit of information dissipates at least $kT \ln 2$ of energy. Change therefore has a floor price, paid in erasure.

Two further results of that program are used here without re-deriving them. Forgetting and recording cost the same: the two are the ends of one reversible edge, and a round trip that records a process and then forgets it returns to the start with no net cost and no preferred direction — an *equal-cost* law with no bias and no arrow. And forgetting is not deletion: a change is real only if it leaves a measurable, equal-and-opposite record — a *shadow* — and erasing that record means traversing it in reverse back to its origin, not zeroing it, because a record dropped without traversal leaves the change unmeasurable, hence unproven. These are the tools; the question of this paper is what sets the *size* of the price.

2. The change budget

Under C₁, change is the sole currency, and a moving body must spend a fixed budget of it. The relativistic statement is that the four-velocity has constant magnitude: a body always moves through spacetime at the speed of light, dividing that fixed speed between motion through space and motion through time. Writing $\beta = v/c$ for the fraction spent on spatial motion, the split is exact and Pythagorean,

$$v_{\text{space}}^2 + v_{\text{time}}^2 = c^2, \quad \text{with } v_{\text{space}} = \beta c, \quad v_{\text{time}} = c \cdot \sqrt{1-\beta^2}.$$

The temporal term $v_{\text{time}} = c\sqrt{1-\beta^2}$ is the rate at which the body advances through its own time — its proper-time rate, scaled by c . It is exactly the budget that spatial motion did not consume. We confirmed the identity across the range of β :

beta	v_space	v_time (internal)	v_space^2 + v_time^2
0.00	0.00000	1.000000	1.000000000000
0.50	0.50000	0.866025	1.000000000000
0.866	0.86603	0.500001	1.000000000000
0.99	0.99000	0.141067	1.000000000000
0.9999	0.99990	0.014142	1.000000000000
max deviation from c^2 : 1.11e-16 (exact)			

The reading C₁ supplies is that v_{time} is *internal change*: the amount a process alters within itself, as distinct from its displacement through space. A body at rest spends the entire budget internally; a body at light speed spends none. What accrues internally is proper time, and proper time is the accumulated internal change — the two are one quantity seen from physics and from the axiom respectively.

3. The cost of forgetting is proper time

A process leaves a record whose size is the number of distinct internal states it passed through. That count is its proper time measured in elementary internal steps: more proper time, more states, more record. To forget the process is to erase that record, and by Landauer erasing it costs in proportion to the number of states erased. Composing the two statements gives the result of this paper:

The cost to forget a process equals its accrued proper time. Per unit of external (coordinate) time, that cost is $\sqrt{1-\beta^2}$.

The two limits are immediate and match intuition. A process at light speed ($\beta \rightarrow 1$) accrues no proper time, passes through no internal states, and leaves nothing to erase: it costs essentially nothing to forget. A process at rest ($\beta \rightarrow 0$) accrues the maximum proper time, passes through every internal state available to it, and costs the most. Speed trades external motion for internal change, and it is the internal change — the record — that must be paid to erase. A photon carries no forgettable history; a body at rest carries all of it.

This also locates the phenomenon in a single physical quantity rather than an abstract notion of “data.” The record is not arbitrary information that happens to accumulate; it is proper time, the same clock that governs aging and decay. What is expensive to forget is exactly what has lived the most internal time.

4. The law is bounded, and it plateaus

A natural first guess is that the cost of forgetting scales inversely with speed — fast is cheap, slow is dear, without bound. C1 corrects this. Because the cost is proper time, it is not $1/\beta$ but $\sqrt{1-\beta^2}$: a bounded function that equals one at rest and falls to zero only at light speed, and — crucially — one that stays near its maximum across the entire slow regime.

β (fraction of c)	cost to forget $\sqrt{1-\beta^2}$	reading
0.00 (rest)	1.000000	maximum — the full record
0.10	0.994987	still ~maximum — the plateau
0.50	0.866025	barely cheaper
0.75	0.661438	—
$0.866025 = \sqrt{3}/2$	0.500000	first half-price point
0.99	0.141067	—
0.9999	0.014142	near light speed — almost free

The consequence is concrete: forgetting does not become even half price until a process moves at 0.866 c. Everything slower sits high on the plateau, at nearly the maximum cost. Ordinary experience — and any process bound to matter at everyday speeds — is therefore uniformly expensive to forget; the discount is a relativistic effect reserved for the near-light-speed limit, not a gradual saving available at any pace. The naive inverse-speed picture predicts cheap forgetting for anything merely fast; the bounded law says that only the truly relativistic escape the plateau.

5. The chord and the arc

The companion paper shows that the path connecting two states is either a line — a direct jump, the chord, an address resolved without visiting anything between — or a wave — the arc, a traversal that realizes every

intermediate state. The cost of forgetting is exactly the choice between them. We counted the internal states an episode of fixed external duration realizes as a function of speed:

beta	states realized in a 1000-tick episode	path
0.00000	1000	full ARC (every state)
0.50000	866	
0.90000	436	
0.99000	141	
0.99999	4	the CHORD (one jump)

At rest the process traverses the full arc and realizes every state; forgetting means erasing all of them, the maximum. Near light speed it collapses toward the chord — a single jump with no intermediate states — and there is almost nothing to erase. Light speed *is* the chord, the invariant line, the direct address; rest is the full arc, the wave that visits everything. The cost of forgetting is the length of the arc actually taken.

The word “almost” is load-bearing. The count never falls below one: even the fastest real process realizes at least its own endpoint, and C_1 's ground-state condition — no state is ever perfectly still, $\langle p^2 \rangle > 0$ — guarantees a minimum of internal jitter that cannot be spent away. A true photon, at exactly c , takes the chord exactly and costs exactly nothing; anything with mass approaches that limit without reaching it, and so is almost, but never exactly, free to forget.

6. Equal cost, and discharge rather than deletion

Because forgetting costs proper time, and by the equal-cost law recording costs the same, a high-proper-time episode is expensive on both ends: dear to lay down and equally dear to remove. The two are one edge traversed in opposite directions, and a full round trip — record forward, forget backward — nets to zero, with no bias between the directions:

episode at rest (maximal internal change): 1000 states		
cost to record (lay the shadow)	:	+1000
cost to forget (walk the shadow back):	:	-1000
round-trip net	:	0 (balanced, no arrow)

The reverse direction is the whole of forgetting. A record cannot be zeroed; it can only be discharged by traversing it back to its origin, one state at a time, because a record dropped without traversal leaves the change it recorded unmeasurable — and an unmeasurable change is, by C_1 , an unproven one. The engine rejects the shortcut: an attempt to delete a thousand-state record without walking it back fails as illegal.

This gives a precise account of persistence. A process that accrued a great deal of proper time — an episode lived slowly and in full internal detail — leaves a correspondingly large record, and by the equal-cost law that record can be discharged only at the cost it took to create, and only by reverse traversal. It cannot be deleted; it can only be walked back. Persistence is not a failure of forgetting but its exact price: a large record simply *is* expensive to discharge, and the sole route to discharging it is to traverse the shadow to its origin. The clinical

shape of this — that intrusive, high-detail episodes resist erasure and are relieved not by suppression but by being led back, step by step, to their source — is the same statement read in a human register, though we make no clinical claim and leave that mapping to the relevant field.

7. Scope and status

Standard, not claimed as new. The constancy of the four-velocity's magnitude, the Pythagorean split of the speed through spacetime, and proper time τ with $d\tau/dt = \sqrt{1-\beta^2}$ are textbook special relativity (Einstein, 1905; Minkowski, 1909). Landauer's bound and the logical reversibility of computation (Landauer, 1961; Bennett, 1973) are established. None of this is original here, and the numerical checks in Sections 2–5 merely confirm the identities.

The contribution. What we add is a set of identifications, each licensed by the no-fixed-point axiom and its cascade: that internal change and proper time are the same quantity; that the cost of forgetting a process equals its accrued proper time; that this cost is the chord–arc length of the companion result, bounded and plateaued rather than inversely proportional to speed; and that forgetting is a reverse traversal of a measurable record, so that persistence is an undischarged traversal rather than a failure to delete. These are interpretive claims built on verified identities; they stand on the coherence of the axiom and the cascade, and they are stated so that they can be contested.

Not claimed. We do not assert a mechanism by which biological memory implements proper-time accounting, and the reference to intrusive-memory phenomena in Section 6 is an analogy in the axiom's register, not a clinical result. The quantitative law $\sqrt{1-\beta^2}$ is a statement about a process's internal time and its erasure cost; its application to any particular physical or biological substrate would require that substrate's own model.

8. Coda

Change costs erasure; erasure costs proper time; proper time is the internal change a thing has undergone. A photon, spending all of its budget on motion, undergoes none and costs nothing to forget — it takes the chord and leaves no history. A body at rest spends everything internally, lives the full arc, and is the most expensive thing there is to undo. Between them the price is bounded and stays high until the very edge of light speed, so that everything slow is dear to forget, and dear in exactly the measure that it has lived. And because the record is a measurable shadow, forgetting is never a deletion but a walk back to the source, paid at the price of the going. What has lived the most internal time is the hardest to let go of — not by accident, but by the same accounting that makes change cost anything at all.

Appendix. Computational summary

All figures were produced in Python with exact floating-point evaluation of the closed forms; no simulation was required. Setting $c = 1$: (A) for β on a grid, $v_{\text{space}} = \beta$ and $v_{\text{time}} = \sqrt{1-\beta^2}$ were formed and $v_{\text{space}}^2 +$

v_{time}^2 compared to 1, with maximum deviation 1.11×10^{-16} . (B) The forget-cost law $f(\beta) = \sqrt{1-\beta^2}$ was tabulated; the half-cost speed solves $\sqrt{1-\beta^2} = \frac{1}{2}$, giving $\beta = \sqrt{3}/2 = 0.866025$, at which $f = 0.500000$ to six places. (C) States realized in an episode of $T = 1000$ coordinate ticks were taken as $\lfloor \sqrt{1-\beta^2} \cdot T \rfloor$ with a floor of one, yielding 1000 at rest and 4 at $\beta = 0.99999$. (D) The equal-cost round trip recorded and then forgot a 1000-state episode for a net of zero, and an attempt to drop the record without reverse traversal was rejected as illegal.

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